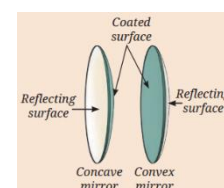


Chapter – 10: Light: Mirrors and Lenses

- Summer holidays – Meena – visit science centre
- Centre – fascinating displays – nature, space, tech.
- Parents – explore section – save water, elec. – Meena, brother – wander off – look around
- One corner – Meena notice – unusual curved mirrors – curious – stepped closer – look into
- Face – unusually large – brother – standing farther – upside down
- Another mirror – tiny version – herself – Meena – puzzled
- Remember act. – mirror – earlier – img. – same size, erect – obj. – chapter – light – curiosity – grade 7
- Seeing confusion – guide – science centre – walked up – says – these mirrors – not plane
- Guide explain – mirrors – spherical – curved inward, outward – img. – diff. look
- Meena's curiosity – grew – decide – talk – teachers – spherical mirrors

What Are Spherical Mirrors?

- Activity –
 - Take – shiny metal spoon – hold – curved surface – close – face
 - Notice – img. – face – diff.? – plane mirror
 - **Observe** img. – move spoon away – obs. Changes
 - Flip spoon – repeat steps
- Notice – spoon – act as mirror – img. – visible
- Look – inner side – spoon – curved inwards – img. – inverted
- Look – outer side – spoon – curved outwards – img. – erect – BUT – smaller
- Curved mirrors – specially made – common type – spherical mirrors – shape – hollow glass sphere
- Mirrors – reflecting surface – spherical – **spherical mirrors**
- Reflecting surface – spherical mirror – curved inward, outward
- Spherical mirror – reflecting surface – curve inwards – **concave mirror** – outline – surface – circular
- Spherical mirror – reflecting surface – curve outwards – **convex mirror**
- Both mirrors – representation – non-reflecting surface – shaded
- **A Step Further**
 - Shape – spherical mirror – such that – part of – imaginary hollow sphere
 - BUT – spherical mirrors – not made – slicing sphere – instead – grinding, polishing – flat glass piece to curved surface
 - Reflective coating – thin layer – aluminium – applied – outer surface – concave mirror – inner surface – convex mirror
- Activity –
 - Place – convex, concave mirrors – table – reflecting surface – upwards
 - View – side – eye level – **identify** – reflecting surface – curve – in, out



What Are the Characteristics of Images Formed by Spherical Mirrors?

- Activity –
 - Take – concave mirror – convex mirror – 2 wooden blocks – place mirror – upright – any obj.

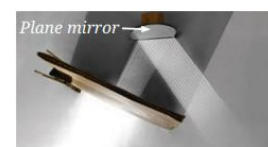
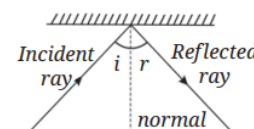
- Place mirrors – adj. – upright – table – keep obj. – front – small dist. – 3-4 cm – check img. – size, erect or not, lateral inversion – record obs.
- Move obj. – away – obs. changes – size, erect or not – record obs.
- Repeat steps – each mirror – **analyse** obs. – **draw** – conclusions
- Concave mirror – obj. – close – mirror – img. – erect, larger – **enlarged**
- Obj. – move away – img. – inverted
- Initially – img. – enlarged – BUT – gets smaller
- Convex mirror – img. – always erect, smaller – **diminished**
- Size – img. – decrease slightly – obj. – move away
- This act. – show – spherical mirrors – diff. behaviour – plane mirrors
- Plane mirror – img. – always erect, same size
- BUT – concave, convex mirror – img. size – vary – obj. dist. – vary
- Also – concave mirror – img. – inverted – obj. – move away
- Lateral inversion – visible – all 3 mirrors
- Reflectors – torches, headlights – cars, scooters – concave
- Dental mirror – dentists – inspect teeth – concave – enlarged view – teeth – held close
- **A Step Further**
 - Remember – learning use – telescope – chapter – beyond Earth – curiosity – grade 6
 - Most modern telescope – curved mirrors – main mirror – large concave mirror
- Side view mirrors – vehicles – convex – img. – always erect, smaller
- Also – convex mirror – curved outside – much wider area
- Convex mirrors – installed – intersections, sharp bends – provide drivers – both sides – visibility – other side – prevent collision
- Convex mirrors – installed – big stores – monitor large areas – deter thefts

What Are the Laws of Reflection?

- Repeat act. – earlier – grade 7 – this time – extend further
- Activity –
 - Collect – plane mirror – stand, torch, comb, paper clip, sheet – white, black paper
 - Did earlier – make thin slit – cover openings – comb – black paper – except one
 - Spread sheet – white paper – table – place mirror – upright
 - Use thin slit, torch – obtain thin beam – light – adjust – fall – mirror
 - Move slit, torch – light beam – fall – diff. angle – reflected beam – shift
 - Make light beam – fall – mirror – diff. angles – obs. – dir. – reflected beam
- Understand better – **draw** on paper – before that – learn – represent light
- Often – represent light – st. lines – arrows – **rays** – indicate path – light travels
 - Draw line – show position – plane mirror – also draw rays – indicate light beam, reflected mirror
- Light ray – fall – mirror – **incident ray** – light ray – comes back – **reflected ray**
 - Remove mirror – pt. – incident ray – strike mirror – draw line – 90° to mirror – this line – **normal** – reflecting surface – incidence pt. O
- Angle – normal, incident ray – **angle of incidence** (i) – angle – normal, reflected ray – **angle of reflection** (r)
 - Your drawing – **measure** – incidence angle, reflection angle – note



- Repeat activity – several times – change incidence angle
- Finally – incident beam – fall – mirror – normal – obs. dir. – reflected beam – both angles – 0 – this case
- Notice – both angles – nearly equal – exp. shows – **angle of incidence (i) = angle of reflection (r)** – law of reflection
- Activity –
 - Same setup – previous act. – BUT – place – stiff sheet – chart paper – flat – table – some part -extend beyond edge
 - Shine – light beam – mirror – obs. – reflected beam – extended portion
 - Bend – extended portion – reflected beam – still visible?
 - Flatten paper – obs.
- Reflected beam – disappear – sheet – bent – reappear – flattened again
- This show – reflected beam – same plane – incident beam – bent sheet – new plane – break alignment
- **Incident ray, normal at point of incidence, reflected ray – same plane** – another law of reflection
- **A Step Further**
 - 2 cases – dir. – incident ray – diff. – fall – same pt. – mirror – BUT – dir. – normal – same
 - Dir. – reflected ray – such that – incident ray, normal, reflected ray – same plane – both cases
- Laws of reflection – valid – mirrors – all kinds – plane, spherical
- BUT – multiple parallel rays – fall – spherical mirror – interesting obs.
- Activity –
 - Collect – plane mirror, concave mirror, convex mirror, stands, torch, comb, paper clip
 - Same setup – above act. – BUT – multiple slits – obtain – multiple parallel beams
 - Multiple beams – fall – plane, concave, convex mirror – obs. – reflected beams
- Multiple parallel beams – fall – plane mirror – multiple reflected beams – parallel
- Concave – reflected beams – get close – **converge** – BUT – convex – reflected beams – spread – **diverge**
- **A Step Further**
 - Draw – obs. – above act. – fig. – shown here
- Activity –
 - Take – concave mirror, sheet – thin paper, newspaper
 - Hold – concave mirror – reflecting surface – face sun – direct – reflected light – paper sheet
 - Adjust dist. – paper – until sharp bright spot
 - Hold mirror – paper sheet – steady – few minutes – paper burn
- Bright spot – paper – reason – sun light – after reflection – conc. – this pt. – produce – sufficient heat – ignite paper
- **A Step Further**
 - Devices – conc. – sunlight – small area – use mirrors, lens – solar concentrators
 - Conc. Sunlight – used – heat liquid – produce steam – generate elec. – OR – heat – various purposes – large scale cooking, solar furnace
 - Solar furnace – melt steel



What Is a Lens?

- Imagine – look through – flat transparent glass window – all obj. – look same – size, shape
- Activity –
 - Collect – flat strip – glass, clear plastic, oil, dropper, water, some text
 - Spread oil – surface – glass, plastic – rub – thin coating – can also use wax
 - Use dropper, finger – place – small drop – water – oiled, waxed spot
 - **Examine** – water drop – shape, curve – inward, outward?
 - Place paper – underneath – glass, plastic – text – directly below water drop
 - Look down – water drop – text below – obs. – change – letters – enlarged, smaller
- Surface – water drop – curved outside – letters under drop – look diff. – appear larger
- Curved surface – water drop – act like – simple lens
- Lens – help reading – small print – make letters appear larger
- **Lens** – piece – transparent material – made of – glass, plastic – curved surfaces – concave, convex
- Lens – thicker – middle – compared – edges – **convex lens**
- Lens – thicker – edges – compared – middle – **concave lens**
- Lens allow light – pass through – things visible – through lens – instead – inside them
- Activity –
 - Collect – convex lens, concave lens, lens holder, small obj.
 - Take – convex lens – place upright – use holder
 - Place obj. – behind lens
 - Look – obj. – lens – other side – note obs. – notebook
 - Move obj. – away – obs. – changes – img.
 - Repeat steps – concave lens
 - Analyse obs. – **compare** – img. – both lenses
- Obj. – convex lens – small dist. – obj. – erect, enlarged
- Dist. – increase – obj. – appear inverted
- Initially – enlarged – then – diminished
- Obj. – concave lens – always – erect, diminished
- Size change – dist. – increase
- Activity –
 - Collect – thin transparent glass plate, convex lens, concave lens, torch, comb, paper clip, 2 books, white sheet paper
 - Use 2 books – adj. – fix glass plate, lens – upright – spread sheet – both books
 - Multiple parallel beams – fall – glass plate, lens – by turns
 - **Record** – analyse – obs.
- Light beam – pass through – glass plate
- Convex lens – converge – light beam – **converging lens**
- Concave lens – diverge – light beam – **diverging lens**
- **A Step Further**
 - Draw – obs. – above act. – fig. – shown here
- Activity –
 - Repeat act. – concave mirror – sun – use – convex lens
- Lens – imp. – used everywhere – eyeglasses, cameras, telescopes, microscopes
- Our eye – convex lens – amazing lens – change shape – allow us – read book, view distant obj.

